Comprehensive Multi-Mode Kinematic Analysis

Python Solver

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1 DH Matrices

Link 1

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 1.0L\cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 1.0L\sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 2

$$\begin{bmatrix} \cos(\theta_2) & 0 & -\sin(\theta_2) & 2.0L\cos(\theta_2) \\ \sin(\theta_2) & 0 & \cos(\theta_2) & 2.0L\sin(\theta_2) \\ 0 & -1 & 0 & 5.0L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 3

$$\begin{bmatrix} \cos(\theta_3 + 1.0\pi) & 0 & \sin(\theta_3 + 1.0\pi) & 0 \\ \sin(\theta_3 + 1.0\pi) & 0 & -\cos(\theta_3 + 1.0\pi) & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 4

$$\begin{bmatrix} \cos(\theta_4) & 0 & -\sin(\theta_4) & 2.5L\cos(\theta_4) \\ \sin(\theta_4) & 0 & \cos(\theta_4) & 2.5L\sin(\theta_4) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 5

$$\begin{bmatrix} \cos(\theta_5) & 0 & \sin(\theta_5) & 2.5L\cos(\theta_5) \\ \sin(\theta_5) & 0 & -\cos(\theta_5) & 2.5L\sin(\theta_5) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 6

$$\begin{bmatrix} \cos(\theta_6 + 1.0\pi) & 0 & -\sin(\theta_6 + 1.0\pi) & 0 \\ \sin(\theta_6 + 1.0\pi) & 0 & \cos(\theta_6 + 1.0\pi) & 0 \\ 0 & -1 & 0 & 2.5L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Link 7

$$\begin{bmatrix} \cos(\theta_7) & 0 & -\sin(\theta_7) & 2.0L\cos(\theta_7) \\ \sin(\theta_7) & 0 & \cos(\theta_7) & 2.0L\sin(\theta_7) \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2 Loop-Closure Equations

2.1 Set 1: $T_1T_2T_3T_4 = T_7^{-1}T_6^{-1}T_5^{-1}$

$$\begin{aligned}
\text{Permutation 1, Element (n˙x)} : & -\sin(\theta_1)\sin(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi) - \sin(\theta_1)\sin(\theta_4)\cos(\theta_2) \\
& - \sin(\theta_2)\sin(\theta_4)\cos(\theta_1) + \frac{\sin^2(\theta_5)\sin^2(\theta_7)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& - \frac{\sin^2(\theta_5)\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} - \frac{\sin^2(\theta_5)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& + \frac{\sin^2(\theta_5)}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} + \sin(\theta_5)\sin(\theta_7) \\
& - \frac{\sin^2(\theta_7)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} + \frac{\sin^2(\theta_7)}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& + \frac{\sin^2(\theta_6+1.0\pi)}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} + \cos(\theta_1)\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi) \\
& - \frac{1}{\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
= & \ 0
\end{aligned}$$

$$\begin{aligned}
\text{Permutation 1, Element (o˙x)} : & \sin(\theta_1)\sin(\theta_2)\sin(\theta_3+1.0\pi) - \frac{\sin(\theta_5)\sin^2(\theta_7)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& + \frac{\sin(\theta_5)\sin^2(\theta_7)}{\cos(\theta_7)\cos(\theta_6+1.0\pi)} + \frac{\sin(\theta_5)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& - \frac{\sin(\theta_5)}{\cos(\theta_7)\cos(\theta_6+1.0\pi)} - \sin(\theta_7)\cos(\theta_5) - \sin(\theta_3+1.0\pi)\cos(\theta_1)\cos(\theta_2) \\
= & \ 0
\end{aligned}$$

$$\begin{aligned}
\text{Permutation 1, Element (a˙x)} : & \sin(\theta_1)\sin(\theta_2)\sin(\theta_4)\cos(\theta_3+1.0\pi) - \sin(\theta_1)\cos(\theta_2)\cos(\theta_4) \\
& - \sin(\theta_2)\cos(\theta_1)\cos(\theta_4) - \sin(\theta_4)\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi) \\
& + \frac{\sin^2(\theta_7)\sin(\theta_6+1.0\pi)}{\cos(\theta_7)} - \frac{\sin(\theta_6+1.0\pi)}{\cos(\theta_7)} \\
= & \ 0
\end{aligned}$$

$$\begin{aligned}
\text{Permutation 1, Element (p˙x)} : & -L\,(2.5\sin(\theta_4)\sin(\theta_1+\theta_2) + 2.5\sin(\theta_7) - 1.0\cos(\theta_1) \\
& + 2.5\cos(\theta_3)\cos(\theta_4)\cos(\theta_1+\theta_2) + 2.5\cos(\theta_6)\cos(\theta_7) - 2.0\cos(\theta_1+\theta_2) - 2.0) \\
= & \ 0
\end{aligned}$$

2.2 Set 2: $T_2T_3T_4T_5 = T_1^{-1}T_7^{-1}T_6^{-1}$

$$\begin{aligned}
\text{Permutation 2, Element (n˙x)} : & \frac{\sin^2(\theta_1)\sin^2(\theta_7)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} - \frac{\sin^2(\theta_1)\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& - \frac{\sin^2(\theta_1)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} + \frac{\sin^2(\theta_1)}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& - \sin(\theta_1)\sin(\theta_6+1.0\pi) - \sin(\theta_2)\sin(\theta_4)\cos(\theta_5) \\
& - \sin(\theta_5)\sin(\theta_3+1.0\pi)\cos(\theta_2) - \frac{\sin^2(\theta_7)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& + \frac{\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} + \frac{\sin^2(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
& + \cos(\theta_2)\cos(\theta_4)\cos(\theta_5)\cos(\theta_3+1.0\pi) - \frac{1}{\cos(\theta_1)\cos(\theta_7)\cos(\theta_6+1.0\pi)} \\
= & \ 0
\end{aligned}$$

$$\begin{aligned}\text{Permutation 2, Element (o˙x)} : & -\frac{\sin^2(\theta_1)\sin^2(\theta_7)\sin(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)}+\frac{\sin^2(\theta_1)\sin(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)}\\ & +\sin(\theta_1)\cos(\theta_6+1.0\pi)-\sin(\theta_2)\cos(\theta_4)-\sin(\theta_4)\cos(\theta_2)\cos(\theta_3+1.0\pi)\\ & +\frac{\sin^2(\theta_7)\sin(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)}-\frac{\sin(\theta_6+1.0\pi)}{\cos(\theta_1)\cos(\theta_7)}\\ & =0\end{aligned}$$

$$\begin{aligned}\text{Permutation 2, Element (a˙x)} : & -\frac{\sin^2(\theta_1)\sin(\theta_7)}{\cos(\theta_1)}-\sin(\theta_2)\sin(\theta_4)\sin(\theta_5)\\ & +\sin(\theta_5)\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)+\frac{\sin(\theta_7)}{\cos(\theta_1)}\\ & +\sin(\theta_3+1.0\pi)\cos(\theta_2)\cos(\theta_5)\\ & =0\end{aligned}$$

$$\begin{aligned}\text{Permutation 2, Element (p˙x)} : & L\,(-2.5\sin(\theta_2)\sin(\theta_4)\cos(\theta_5)-2.5\sin(\theta_2)\sin(\theta_4)+2.5\sin(\theta_3)\sin(\theta_5)\cos(\theta_2)\\ & \qquad -2.5\sin(\theta_7)\cos(\theta_1)+2.0\cos(\theta_1)-2.5\cos(\theta_2)\cos(\theta_3)\cos(\theta_4)\cos(\theta_5)\\ & \qquad -2.5\cos(\theta_2)\cos(\theta_3)\cos(\theta_4)+2.0\cos(\theta_2)+1.0)\\ & =0\end{aligned}$$

2.3 Set 3: $T_3T_4T_5T_6 = T_2^{-1}T_1^{-1}T_7^{-1}$

$$\begin{aligned}\text{Permutation 3, Element (n˙x)} : & \frac{\sin^2(\theta_1)\sin^2(\theta_2)\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}-\frac{\sin^2(\theta_1)\sin^2(\theta_2)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}\\ & -\frac{\sin^2(\theta_1)\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}+\frac{\sin^2(\theta_1)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}\\ & -\frac{\sin(\theta_1)\sin(\theta_2)\sin^2(\theta_7)}{\cos(\theta_7)}+\frac{\sin(\theta_1)\sin(\theta_2)}{\cos(\theta_7)}-\frac{\sin^2(\theta_2)\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}\\ & +\frac{\sin^2(\theta_2)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}-\sin(\theta_4)\sin(\theta_6+1.0\pi)\cos(\theta_3+1.0\pi)\\ & -\sin(\theta_5)\sin(\theta_3+1.0\pi)\cos(\theta_6+1.0\pi)+\frac{\sin^2(\theta_7)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}\\ & +\cos(\theta_4)\cos(\theta_5)\cos(\theta_3+1.0\pi)\cos(\theta_6+1.0\pi)-\frac{1}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_7)}\\ & =0\end{aligned}$$

$$\begin{aligned}\text{Permutation 3, Element (o˙x)} : & -\frac{\sin^2(\theta_1)\sin^2(\theta_2)\sin(\theta_7)}{\cos(\theta_1)\cos(\theta_2)}+\frac{\sin^2(\theta_1)\sin(\theta_7)}{\cos(\theta_1)\cos(\theta_2)}+\sin(\theta_1)\sin(\theta_2)\sin(\theta_7)\\ & +\frac{\sin^2(\theta_2)\sin(\theta_7)}{\cos(\theta_1)\cos(\theta_2)}-\sin(\theta_5)\cos(\theta_4)\cos(\theta_3+1.0\pi)-\frac{\sin(\theta_7)}{\cos(\theta_1)\cos(\theta_2)}\\ & -\sin(\theta_3+1.0\pi)\cos(\theta_5)\\ & =0\end{aligned}$$

$$\begin{aligned}\text{Permutation 3, Element (a˙x)} : & -\frac{\sin(\theta_1)\sin^2(\theta_2)}{\cos(\theta_2)}+\frac{\sin(\theta_1)}{\cos(\theta_2)}+\sin(\theta_2)\cos(\theta_1)\\ & -\sin(\theta_4)\cos(\theta_3+1.0\pi)\cos(\theta_6+1.0\pi)+\sin(\theta_5)\sin(\theta_3+1.0\pi)\sin(\theta_6+1.0\pi)\\ & -\sin(\theta_6+1.0\pi)\cos(\theta_4)\cos(\theta_5)\cos(\theta_3+1.0\pi)\\ & =0\end{aligned}$$

$$\begin{aligned}\text{Permutation 3, Element (p˙x)} : & L\left(-2.0\sin(\theta_1)\sin(\theta_2)-2.5\sqrt{2}\sin(\theta_3)\cos\left(\theta_5+\frac{\pi}{4}\right)\right.\\ & \qquad -2.5\sqrt{2}\sin\left(\theta_5+\frac{\pi}{4}\right)\cos(\theta_3)\cos(\theta_4)+2.0\cos(\theta_1)\cos(\theta_2)+1.0\cos(\theta_2)\\ & \qquad \left.-2.5\cos(\theta_3)\cos(\theta_4)+2.0\right)\\ & =0\end{aligned}$$

2.4 Set 4: $T_4T_5T_6T_7 = T_3^{-1}T_2^{-1}T_1^{-1}$

$$
\begin{aligned}
\text{Permutation 4, Element (n˙x)} : & \frac{\sin^2(\theta_1)\sin^2(\theta_2)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} - \frac{\sin^2(\theta_1)\sin^2(\theta_2)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} \\
& - \frac{\sin^2(\theta_1)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} + \frac{\sin^2(\theta_1)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} \\
& - \frac{\sin(\theta_1)\sin(\theta_2)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_3+1.0\pi)} + \frac{\sin(\theta_1)\sin(\theta_2)}{\cos(\theta_3+1.0\pi)} \\
& - \frac{\sin^2(\theta_2)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} + \frac{\sin^2(\theta_2)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} \\
& - \sin(\theta_4)\sin(\theta_6+1.0\pi)\cos(\theta_7) - \sin(\theta_5)\sin(\theta_7)\cos(\theta_4) \\
& + \frac{\sin^2(\theta_3+1.0\pi)}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} + \cos(\theta_4)\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi) \\
& - \frac{1}{\cos(\theta_1)\cos(\theta_2)\cos(\theta_3+1.0\pi)} \\
= & \ 0 \\
\text{Permutation 4, Element (o˙x)} : & -\frac{\sin(\theta_1)\sin^2(\theta_2)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_2)\cos(\theta_3+1.0\pi)} + \frac{\sin(\theta_1)\sin^2(\theta_2)}{\cos(\theta_2)\cos(\theta_3+1.0\pi)} \\
& + \frac{\sin(\theta_1)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_2)\cos(\theta_3+1.0\pi)} - \frac{\sin(\theta_1)}{\cos(\theta_2)\cos(\theta_3+1.0\pi)} \\
& + \frac{\sin(\theta_2)\sin^2(\theta_3+1.0\pi)\cos(\theta_1)}{\cos(\theta_3+1.0\pi)} - \frac{\sin(\theta_2)\cos(\theta_1)}{\cos(\theta_3+1.0\pi)} + \sin(\theta_4)\cos(\theta_6+1.0\pi) \\
& + \sin(\theta_6+1.0\pi)\cos(\theta_4)\cos(\theta_5) \\
= & \ 0 \\
\text{Permutation 4, Element (a˙x)} : & \sin(\theta_4)\sin(\theta_7)\sin(\theta_6+1.0\pi) - \sin(\theta_5)\cos(\theta_4)\cos(\theta_7) \\
& - \sin(\theta_7)\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi) + \sin(\theta_3+1.0\pi) \\
= & \ 0 \\
\text{Permutation 4, Element (p˙x)} : & L\Big(5.0\sin(\theta_3) + 2.0\sin(\theta_4)\sin(\theta_6)\cos(\theta_7) - 2.0\sin(\theta_5)\sin(\theta_7)\cos(\theta_4) \\
& + 2.5\sqrt{2}\sin\Big(\theta_5+\frac{\pi}{4}\Big)\cos(\theta_4) - 1.0\cos(\theta_2)\cos(\theta_3) - 2.0\cos(\theta_3) \\
& - 2.0\cos(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) + 2.5\cos(\theta_4)\Big) \\
= & \ 0
\end{aligned}
$$

2.5 Set 5: $T_5T_6T_7T_1 = T_4^{-1}T_3^{-1}T_2^{-1}$

$$
\begin{aligned}
\text{Permutation 5, Element (n˙x)} : & \sin(\theta_1)\sin(\theta_6+1.0\pi)\cos(\theta_5) + \frac{\sin^2(\theta_2)\sin^2(\theta_4)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} \\
& - \frac{\sin^2(\theta_2)\sin^2(\theta_4)}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} - \frac{\sin^2(\theta_2)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} \\
& + \frac{\sin^2(\theta_2)}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} + \sin(\theta_2)\sin(\theta_4) \\
& - \frac{\sin^2(\theta_4)\sin^2(\theta_3+1.0\pi)}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} + \frac{\sin^2(\theta_4)}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} \\
& - \sin(\theta_5)\sin(\theta_7)\cos(\theta_1) + \frac{\sin^2(\theta_3+1.0\pi)}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} \\
& + \cos(\theta_1)\cos(\theta_5)\cos(\theta_7)\cos(\theta_6+1.0\pi) - \frac{1}{\cos(\theta_2)\cos(\theta_4)\cos(\theta_3+1.0\pi)} \\
= & \ 0
\end{aligned}
$$

$$\begin{aligned}\text{Permutation 5, Element (o}\dot{\text{x}}\text{)} : &\sin(\theta_1)\sin(\theta_5)\sin(\theta_7) - \sin(\theta_1)\cos(\theta_5)\cos(\theta_7)\cos(\theta_6 + 1.0\pi) \\ &- \frac{\sin(\theta_2)\sin^2(\theta_4)\sin^2(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_3 + 1.0\pi)} + \frac{\sin(\theta_2)\sin^2(\theta_4)}{\cos(\theta_4)\cos(\theta_3 + 1.0\pi)} \\ &+ \frac{\sin(\theta_2)\sin^2(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_3 + 1.0\pi)} - \frac{\sin(\theta_2)}{\cos(\theta_4)\cos(\theta_3 + 1.0\pi)} - \sin(\theta_4)\cos(\theta_2) \\ &+ \sin(\theta_6 + 1.0\pi)\cos(\theta_1)\cos(\theta_5) \\ = &\ 0\end{aligned}$$

$$\begin{aligned}\text{Permutation 5, Element (a}\dot{\text{x}}\text{)} : &-\frac{\sin^2(\theta_4)\sin(\theta_3 + 1.0\pi)}{\cos(\theta_4)} - \sin(\theta_5)\cos(\theta_7) \\ &- \sin(\theta_7)\cos(\theta_5)\cos(\theta_6 + 1.0\pi) + \frac{\sin(\theta_3 + 1.0\pi)}{\cos(\theta_4)} \\ = &\ 0\end{aligned}$$

$$\begin{aligned}\text{Permutation 5, Element (p}\dot{\text{x}}\text{)} : &-L\Big(1.0\sin(\theta_1)\sin(\theta_6)\cos(\theta_5) - 5.0\sin(\theta_3)\cos(\theta_4) + 1.0\sin(\theta_5)\sin(\theta_7)\cos(\theta_1) \\ &+ 2.0\sin(\theta_5)\sin(\theta_7) - 2.5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right) + 1.0\cos(\theta_1)\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) \\ &+ 2.0\cos(\theta_3)\cos(\theta_4) + 2.0\cos(\theta_5)\cos(\theta_6)\cos(\theta_7) - 2.5\Big) \\ = &\ 0\end{aligned}$$

2.6 Set 6: $T_6T_7T_1T_2 = T_5^{-1}T_4^{-1}T_3^{-1}$

$$\begin{aligned}\text{Permutation 6, Element (n}\dot{\text{x}}\text{)} : &-\sin(\theta_1)\sin(\theta_2)\cos(\theta_7)\cos(\theta_6 + 1.0\pi) + \sin(\theta_1)\sin(\theta_6 + 1.0\pi)\cos(\theta_2) \\ &+ \sin(\theta_2)\sin(\theta_6 + 1.0\pi)\cos(\theta_1) + \frac{\sin^2(\theta_4)\sin^2(\theta_5)\sin^2(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} \\ &- \frac{\sin^2(\theta_4)\sin^2(\theta_5)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} - \frac{\sin^2(\theta_4)\sin^2(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} \\ &+ \frac{\sin^2(\theta_4)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} - \frac{\sin^2(\theta_5)\sin^2(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} \\ &+ \frac{\sin^2(\theta_5)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} + \sin(\theta_5)\sin(\theta_3 + 1.0\pi) \\ &+ \frac{\sin^2(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} + \cos(\theta_1)\cos(\theta_2)\cos(\theta_7)\cos(\theta_6 + 1.0\pi) \\ &- \frac{1}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_3 + 1.0\pi)} \\ = &\ 0\end{aligned}$$

$$\begin{aligned}\text{Permutation 6, Element (o}\dot{\text{x}}\text{)} : &-\frac{\sin^2(\theta_4)\sin^2(\theta_5)\sin(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)} + \frac{\sin^2(\theta_4)\sin(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)} \\ &+ \frac{\sin^2(\theta_5)\sin(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)} - \sin(\theta_5)\cos(\theta_3 + 1.0\pi) \\ &+ \sin(\theta_7)\cos(\theta_6 + 1.0\pi) - \frac{\sin(\theta_3 + 1.0\pi)}{\cos(\theta_4)\cos(\theta_5)} \\ = &\ 0\end{aligned}$$

$$\begin{aligned}\text{Permutation 6, Element (a}\dot{\text{x}}\text{)} : &-\sin(\theta_1)\sin(\theta_2)\sin(\theta_6 + 1.0\pi) - \sin(\theta_1)\cos(\theta_2)\cos(\theta_7)\cos(\theta_6 + 1.0\pi) \\ &- \sin(\theta_2)\cos(\theta_1)\cos(\theta_7)\cos(\theta_6 + 1.0\pi) + \frac{\sin(\theta_4)\sin^2(\theta_5)}{\cos(\theta_5)} - \frac{\sin(\theta_4)}{\cos(\theta_5)} \\ &+ \sin(\theta_6 + 1.0\pi)\cos(\theta_1)\cos(\theta_2) \\ = &\ 0\end{aligned}$$

$$\begin{aligned}\text{Permutation 6, Element (p}\dot{\text{x}}\text{)} : &L\left(-1.0\sin(\theta_1)\sin(\theta_6) - 2.0\sin(\theta_6)\sin(\theta_1 + \theta_2) + 5.0\sin(\theta_7)\cos(\theta_6)\right. \\ &- 1.0\cos(\theta_1)\cos(\theta_6)\cos(\theta_7) + 2.5\cos(\theta_5) - 2.0\cos(\theta_6)\cos(\theta_7)\cos(\theta_1 + \theta_2) \\ &\left. - 2.0\cos(\theta_6)\cos(\theta_7) + 2.5\right) \\ = &\ 0\end{aligned}$$

2.7 Set 7: $T_7T_1T_2T_3 = T_6^{-1}T_5^{-1}T_4^{-1}$

$$
\begin{aligned}
\text{Permutation 7, Element (n˙x)} : & -\sin(\theta_1)\sin(\theta_2)\cos(\theta_7)\cos(\theta_3+1.0\pi) + \frac{\sin^2(\theta_4)\sin^2(\theta_5)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} \\
& - \frac{\sin^2(\theta_4)\sin^2(\theta_5)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} - \frac{\sin^2(\theta_4)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} \\
& + \frac{\sin^2(\theta_4)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} + \sin(\theta_4)\sin(\theta_6+1.0\pi) \\
& - \frac{\sin^2(\theta_5)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} + \frac{\sin^2(\theta_5)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} \\
& + \sin(\theta_7)\sin(\theta_3+1.0\pi) + \frac{\sin^2(\theta_6+1.0\pi)}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} \\
& + \cos(\theta_1)\cos(\theta_2)\cos(\theta_7)\cos(\theta_3+1.0\pi) - \frac{1}{\cos(\theta_4)\cos(\theta_5)\cos(\theta_6+1.0\pi)} \\
= & \ 0 \\
\text{Permutation 7, Element (o˙x)} : & -\sin(\theta_1)\cos(\theta_2)\cos(\theta_7) - \sin(\theta_2)\cos(\theta_1)\cos(\theta_7) \\
& - \frac{\sin(\theta_4)\sin^2(\theta_5)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_5)\cos(\theta_6+1.0\pi)} + \frac{\sin(\theta_4)\sin^2(\theta_5)}{\cos(\theta_5)\cos(\theta_6+1.0\pi)} \\
& + \frac{\sin(\theta_4)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_5)\cos(\theta_6+1.0\pi)} - \frac{\sin(\theta_4)}{\cos(\theta_5)\cos(\theta_6+1.0\pi)} - \sin(\theta_6+1.0\pi)\cos(\theta_4) \\
= & \ 0 \\
\text{Permutation 7, Element (a˙x)} : & -\sin(\theta_1)\sin(\theta_2)\sin(\theta_3+1.0\pi)\cos(\theta_7) - \frac{\sin(\theta_5)\sin^2(\theta_6+1.0\pi)}{\cos(\theta_6+1.0\pi)} \\
& + \frac{\sin(\theta_5)}{\cos(\theta_6+1.0\pi)} - \sin(\theta_7)\cos(\theta_3+1.0\pi) \\
& + \sin(\theta_3+1.0\pi)\cos(\theta_1)\cos(\theta_2)\cos(\theta_7) \\
= & \ 0 \\
\text{Permutation 7, Element (p˙x)} : & -L\,(5.0\sin(\theta_7) - 1.0\cos(\theta_1)\cos(\theta_7) + 2.5\cos(\theta_5)\cos(\theta_6) + 2.5\cos(\theta_6) \\
& \qquad - 2.0\cos(\theta_7)\cos(\theta_1+\theta_2) - 2.0\cos(\theta_7)) \\
= & \ 0
\end{aligned}
$$

3 Identified Feasible Driving Modes

The symbolic analyzer identified the following combinations of joints that can theoretically drive the mechanism (allowing sequential analytical solution):

- Input(s): θ_3, θ_5
- Input(s): θ_4, θ_5

4 Numerical Validation Curves for Each Mode

The following plots illustrate the input-output relationships for each feasible driving mode identified above. For multi-degree-of-freedom modes, secondary inputs are fixed at zero for visualization.